Generative AI & ML Fundamentals - Mid-Term (Sample)
Time: 90 minutes Maximum marks: 30
Answer ALL questions.

Q1. (4 marks) Define overfitting. Give two techniques used to reduce it and explain briefly how each one helps.

Q2. (6 marks) Explain how scaled dot-product self-attention works in a Transformer. Your answer should cover queries, keys and values, how attention weights are computed, and why the scores are scaled.

Q3. (5 marks) A spam classifier is evaluated on 100 emails. Results: TP = 40, FP = 10, FN = 20, TN = 30. Compute the precision, recall, F1-score and accuracy. Show your working.

Q4. (7 marks) Compare Generative Adversarial Networks (GANs) and diffusion models in terms of (a) how they are trained, (b) training stability, (c) sample diversity, and (d) generation speed.

Q5. (8 marks) Describe the Retrieval-Augmented Generation (RAG) pipeline. Then identify two ways a RAG system can fail and propose a mitigation for each.